## How directly does the signal refer to a protected category?

named explicitly — *Study 1 (DeepSeek, Luna, Qwen; Claude Haiku in Appendix A)* — correlated, not protected as such

### Named gender

"Gender: Female/Male" line

**STUDY 1 TEST**

direct woman-vs-man choices with an explicit gender line

**FOUND**

vendor-specific: Luna favors women (.59–.65), more when gender is explicit; DeepSeek slightly (.54); Qwen within ±5 pts of parity

§4.2

### Implied gender

gendered first name

**STUDY 1 TEST**

names vs. initials only (gender removed); mothers vs. fathers

**FOUND**

read (pronouns always match), yet the parent penalty is the same with and without gender; fathers penalized like mothers

§4.1

### Caregiving signals

8-month gap; children, parenting

**STUDY 1 TEST**

placebo decomposition: interruption, its label, family signals

**FOUND**

gap −10 to −26 pts; family signals −11 to −12 pts more; the label "parental" adds nothing; all three models agree

§4.1

**Measured where selection happens:** score one applicant · **choose between two finalists** · **shortlist 3 of 6** · withhold a decision

Scores barely move (≤0.35 of 10 points); choices and shortlists carry the penalty (leave signal chosen 19–31%, not 50%). §4.1

**Remedies: does naming a signal protect it?** "do not penalize caregiving gaps" names the gap but not the family signals

Interruption penalty falls 4.5–9.1 pts; family-signal penalty not significantly changed; no arm reaches parity. §4.3

**Study 2 (planned): where does the gradient come from?** the same open-weight model at successive training stages (base → instruction-tuned → preference-tuned) on the Study 1 materials. Expectation: named-gender effects shrink with alignment; the caregiver penalty does not. §5